Projective Information Loss and Fibre Erasure in Admissible Non-Injective Projection

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Abstract

We establish a fibre-erasure theorem for the admissible coarse-graining hierarchy of the Cosmochrony spectral programme [4, 1]. The non-injective projection $\Pi : \Omega \to \mathcal{O}$ defines a residual fibre-information functional $I(c; \sigma(\ell))$, where c is a Weil-block fibre label and $\sigma_c(\ell)$ is the BFS capacity profile at coarse-graining depth ℓ. The structural sufficiency hypothesis [H-suff] — that $\sigma(\ell)$ is a sufficient statistic for the hierarchy beyond ℓ — is proved at the level of the BFS rank observable (Proposition 3.6), where the fibre label is erased identically; for the full Born–Infeld capacity it remains conditional on a single preservation step, numerically supported for $q \in \{61, 151, 211\}$ (Section 5.3). Granting [H-suff], the data-processing inequality [10] implies that $I(c; \sigma(\ell))$ is non-increasing in ℓ: fibre information is erased, not created, under admissible coarse-graining. This result is *not* a c-theorem; it does not count effective degrees of freedom. It is a quantitative realisation of the ENI no-go theorem [4]: non-injectivity of Π forces projective information loss, and the data-processing inequality makes this loss monotone and measurable. The inter-sector variance $\mathrm{Var}_c(\sigma_c(\ell))$ serves as a computable proxy; its restriction to SU(3) colour-triplets connects directly to hypothesis [H-color] [9, 8].

Keywords. non-injective projection; fibre erasure; admissible coarse-graining; data-processing inequality; information monotone; BFS capacity; spectral admissibility; projective information loss; Weil representation; Heisenberg group; [H-color]

Contents

1 Introduction

The ENI no-go theorem [4] establishes a fundamental structural fact: genuine emergence is necessarily non-injective. Any surjective map $\Pi : \Omega \to \mathcal{O}$ satisfying the projective-emergent conditions carries a strictly positive projection entropy $S_\Pi > 0$: information about the microscopic representative is irrecoverably lost at the effective level. This is a qualitative result. It identifies the structure of information loss but says nothing about how this loss is distributed across a hierarchy of resolutions.

The Cosmochrony spectral programme [6, 7] provides a concrete realisation of this hierarchy. For each prime q, the BFS capacity profiles $\sigma_c(\ell)$, indexed by Weil-block fibre label c and coarse-graining depth ℓ, track the residual projective capacity of block c as the BFS cascade advances. Numerical campaigns across $q \in \{61, 151, 211, 307\}$ reveal a consistent pattern: inter-sector variance collapses as depth increases, and the profiles converge toward a universal profile σ^* approached in the limit $q \to \infty$ (Section 5.4). Hypothesis [H-color] [9, 8] identifies one specific instance of this convergence: the SU(3) colour-triplet sectors $\{c, \omega c, \omega^2 c\}$ become indistinguishable at the effective level for $q \equiv 1 \pmod 3$.

This paper addresses the following question: can the qualitative ENI no-go and the quantitative spectral convergence be connected by a principled information-theoretic structure? We show that the answer is yes, conditional on the structural sufficiency hypothesis [H-suff]: the residual fibre-information functional $I(c; \sigma(\ell))$ is non-increasing in the coarse-graining depth ℓ, by the data-processing inequality [10]. The result is conditional, not unconditional: [H-suff] is proved at the rank level (Proposition 3.6) and strongly motivated (Remark 3.2); for the full Born–Infeld capacity it is numerically supported for $q \in \{61, 151, 211\}$ (Section 5.3) but not yet proved analytically.

Two structural distinctions are essential before proceeding. First, the coarse-graining axis ℓ is distinct from and orthogonal to the temporal cascade axis n of the PTO paper [5]: see Table 1 and Remark 2.3. Second, as stated in Section 1.1, ENI alone does not imply monotonicity; the hierarchical structure of [H-suff] and [H-sg] is the informational content this paper contributes beyond ENI.

1.1 Contributions and non-claims

This paper establishes, conditional on hypotheses [H-suff] and [H-sg] (Section 2.4):

(i) a monotone information functional $I(c; \sigma(\ell))$ for the BFS coarse-graining hierarchy (Theorem 4.3);

(ii) its non-increase under depth-increment, via the data-processing inequality;

(iii) the identification of $I(c; \sigma(\ell)) = 0$ as the fibre-erasure equilibrium condition (Corollary 4.4).

This paper does *not* establish:

(i) a c-theorem or analogue (no counting of effective degrees of freedom; see open problem [O-2]);

(ii) the full capacity-level hypothesis [H-suff] itself (stated and motivated; numerically supported for $q \in \{61, 151, 211\}$; proved at the rank level in Proposition 3.6, with the pointwise Born–Infeld residue open, see [O-1a′]);

(iii) the full capacity-level semigroup hypothesis [H-sg] (proved at the rank level in Corollary 3.7, and there redundant; the capacity-level residue is open, see [O-1b]).

Strict disclaimer. This result should not be called a c-theorem. It is a *fibre-erasure theorem*: the monotone measures the loss of residual fibre-label information under admissible coarse-graining. A c-theorem-type statement would require an additional counting principle for effective degrees of freedom, which is not established here and is listed as open problem [O-2] in Section 6.

ENI versus monotonicity. ENI [4] establishes that information is structurally lost under non-injective projection, but does not by itself imply any monotone hierarchy over the coarse-graining scale ℓ. The monotonicity of Theorem 4.3 requires the additional hierarchical structure encoded in [H-suff] and [H-sg]: these are the informational content this paper contributes beyond ENI.

1.2 Programme placement

The present paper occupies a hub position in the programme. ENI [4] and Foundation [1] provide the foundational input: the non-injective projection Π, the Weil-fibre realisation, and the projection entropy S_Π. The spectral admissibility programme [6, 7] provides the operational hierarchy: the BFS capacity profiles $\sigma_c(\ell)$ and their numerical campaigns, including the O31–O32 investigation [9, 8]. This paper connects these two branches by translating the ENI projection entropy into the operational functional $I(c;\sigma(\ell))$ and identifying spectral convergence $\sigma_c \to \sigma^*$ as the thermodynamic universal profile (Section 5.4). The PTO paper [5] handles the complementary temporal axis n (Table 1); the two papers are structurally orthogonal and together constitute the information-theoretic architecture of the admissibility cascade.

2 Framework

2.1 Non-injective projection and fibre structure

We recall the setting of [4, 1].

Definition 2.1 (Non-injective projection and fibre)**.** A *projective description* is a surjective map $\Pi : \Omega \to \mathcal{O}$ such that $\mathcal{O}$ carries no microstate-resolving structure independent of Π. The *fibre* over $o \in \mathcal{O}$ is $\Pi^{-1}(o) = \{\, \omega \in \Omega : \Pi(\omega) = o \,\}$. The *projection entropy* is $S_\Pi = \mathbb{E}_o[\,\log |\Pi^{-1}(o)|\,] \geq 0$, with $S_\Pi > 0$ whenever Π is non-injective (ENI Theorem 2.1, [4]).

In the spectral realisation [1], the fibre structure is implemented via the Weil representation V_ρ of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for a prime q. The minimal non-injectivity is the parity involution $c \leftrightarrow q-c$: Weil blocks ρ_c and ρ_{q-c} map to the same effective observable class. The *fibre label* $c \in \{1, \dots, (q-1)/2\}$ indexes blocks within an equivalence class under Π.

2.2 The BFS coarse-graining hierarchy

Definition 2.2 (BFS capacity profile)**.** For a prime q, a fibre label c, and BFS depth $\ell \in \mathbb{N}$, the *BFS capacity profile* $\sigma_c(\ell) \in [0, 1]$ is the normalised residual projective capacity of block c at depth ℓ along the Heisenberg Cayley graph BFS cascade [6, 7]. The *pair observable* is $\sigma_{\mathrm{pair}}(\ell) = \sigma_c(\ell) \cdot \sigma_{q-c}(\ell)$ on conjugate pair $(c, q-c)$.

For a prime q, the Heisenberg group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ carries a Cayley graph whose BFS shells B_ℓ grow as $|B_\ell| \sim \ell^4$ in the Heisenberg regime [6]. The capacity profile $\sigma_c(\ell)$ measures the residual Gram-Schmidt span of Weil block ρ_c not yet exhausted at shell depth ℓ: it starts at $\sigma_c(0) = 1$ and decreases toward 0 as BFS saturation is approached. The depth parameter $\ell \in \mathbb{N}$ is discrete; the coarse-graining semigroup of Section 2.4 therefore acts on a discrete monoid, not a continuous scale.

Axis	Parameter	Role
Coarse-graining / resolution	ℓ (BFS depth/window)	coarse-graining hierarchy; carries $I(c;\sigma(\ell))$; this pap
Thermodynamic / continuum	$q \to \infty$	Limit for σ^*; established numerically (Section 5.4)
Temporal / cascade	n (cascade step)	Projective time; carries $I(n)$ of [5]

Table 1: The three distinct axes of the observable hierarchy. The coarse-graining axis ℓ is the support of this paper. The temporal axis n is the support of PTO [5]. The continuum axis $q \to \infty$ is the thermodynamic limit in which the fixed point σ^* is approached; Section 5.4 establishes this numerically.

2.3 Three-axis separation

Remark 2.3 (Cascade and coarse-graining are orthogonal). The admissible cascade $U_0 \to U_1 \to \cdots$ defines the temporal axis n and is *not* an RG flow. RG-like behaviour is carried by Π and the semigroup $\{G_{\ell' \leftarrow \ell}\}$ on the ℓ-axis. The cascade *enlarges* the admissible space as temporal ordering advances; an RG flow toward the IR *reduces* effective degrees of freedom. These are distinct phenomena on orthogonal axes. The fibre-erasure fixed point σ^* lives on the thermodynamic axis: it is approached as $q \to \infty$ with $\mathrm{O}(q^{-1/2})$ corrections, not as a Wilsonian ℓ-limit at fixed q (Section 5.4).

2.4 The coarse-graining semigroup

Definition 2.4 (Coarse-graining channel)**.** For $\ell' > \ell$, the *coarse-graining channel* $G_{\ell' \leftarrow \ell} : \sigma(\ell) \mapsto \sigma(\ell')$ is the map sending the BFS capacity profile at depth ℓ to the profile at depth ℓ'.

Hypothesis 2.5 ([H-sg]: semigroup property)**.** The coarse-graining channels satisfy

$$G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1} \qquad \text{for all } \ell_1 < \ell_2 < \ell_3. \tag{1}$$

Hypothesis [H-sg] is plausible from the nested structure of BFS shells: the subgraph induced by B_{ℓ_1} is contained in that induced by B_{ℓ_2} for $\ell_1 < \ell_2$, so the transition $\sigma(\ell_1) \to \sigma(\ell_2)$ can in principle be factored through $\sigma(\ell_1)$ alone. Without [H-sg], the data-processing inequality applied step-by-step via [H-suff] yields $I(c;\sigma(\ell+1)) \le I(c;\sigma(\ell))$ for each consecutive pair, but not the uniform statement of Theorem 4.3 for all $\ell' > \ell$. Note that the semigroup acts on the space of observable descriptions, not on the static substrate Ω: composability is a property of the coarse-graining of effective observables. At the level of the rank observable [H-sg] holds and is moreover redundant (Corollary 3.7); only its capacity-level form remains hypothetical here.

2.5 Probability measure on fibre labels

Definition 2.6 (Fibre-label measure)**.** Let μ be a probability measure on $\{c : 1 \le c \le (q-1)/2\}$. Two canonical choices are used:

(i) *Uniform*: $\mu(c) = 2/(q-1)$ for all c.

(ii) *Triplet-uniform* (for $q \equiv 1 \pmod 3$): μ uniform on $\{c, \omega c, \omega^2 c\}$, where ω is a primitive cube root of unity modulo q; used in the [H-color] context [9, 8].

The choice of μ does not affect the monotonicity of $I(c;\sigma(\ell))$ (which holds for any fixed measure by DPI), but affects its value and the interpretation of the fixed point.

3 Structural Sufficiency and the Markov Property

Hypothesis 3.1 ([H-suff]: structural sufficiency)**.** For all $\ell' > \ell$, the coarse-graining channel $G_{\ell' \leftarrow \ell}$ is conditionally independent of the fibre label c given $\sigma(\ell)$:

$$c \longrightarrow \sigma(\ell) \longrightarrow \sigma(\ell'). \tag{2}$$

That is, $\sigma(\ell)$ is a *sufficient statistic* for $\sigma(\ell')$ with respect to the fibre label c. The term "sufficient" is used in the information-theoretic sense: all information in $\sigma(\ell)$ that is predictive of $\sigma(\ell')$ is already fully encoded; no additional knowledge of c reduces the uncertainty in $\sigma(\ell')$ beyond what $\sigma(\ell)$ already captures.

Remark 3.2 (Interpretive status of [H-suff]). Hypothesis [H-suff] is not introduced solely to enable the DPI. It is the informational reformulation of an effective closure property already implicit throughout the spectral admissibility programme: if [H-suff] failed — if BFS dynamics at depth ℓ' re-accessed label information not encoded in $\sigma(\ell)$ — then $\sigma_c(\ell)$ would not constitute stable effective observables under Π, and the convergence $\sigma_c \to \sigma^*$, the plateau structure across $q \in \{61, 151, 211, 307\}$, and the [H-color] campaign [9, 8] would be structurally very difficult to account for. More precisely, [H-suff] is the condition that the BFS observable $\sigma(\ell)$ constitutes a *complete summary* of the fibre structure at resolution ℓ: the projection Π has already collapsed the relevant fibre information at each level of the hierarchy, and no deeper BFS step retrieves it. This is exactly what the non-injectivity of Π (ENI [4]) demands at the effective level: fibre information is not gradually re-injected as coarse-graining proceeds, but is collapsed once and for all by Π. In this sense, [H-suff] makes explicit, in information-theoretic language, a structural commitment already implicit in the programme's use of $\sigma_c(\ell)$ as an effective observable.

Remark 3.3 (Status of [H-suff]). Hypothesis [H-suff] is strongly motivated by Remark 3.2 and is compatible with all known spectral results. A permutation-based Spearman test on BFS capacity data for $q \in \{61, 151, 211\}$ yields no significant c-dependence in step-by-step residuals at any tested depth (Section 5.3), providing numerical evidence for [H-suff]. Analytical proof at the rank level is given in Proposition 3.6; the remaining Born–Infeld gap to the full capacity is open pointwise, see [O-1a′]. At the averaged-exponent level the same direction is supported by the mean universality of the block decay exponent [3] and the effective colour-triplet form [H-color]$_{\mathrm{eff}}$ [8].

Remark 3.4 (Why [H-suff] is load-bearing). If hypothesis [H-suff] fails for some pair (ℓ, ℓ') — that is, if $\sigma(\ell')$ depends on c beyond its dependence on $\sigma(\ell)$ — then no structural information monotone of the form $I(c; \sigma(\ell')) \leq I(c; \sigma(\ell))$ can be guaranteed. If $p(\sigma(\ell') \mid \sigma(\ell), c) \neq p(\sigma(\ell') \mid \sigma(\ell))$ for some c, then the chain $c \to \sigma(\ell) \to \sigma(\ell')$ is not Markov, and the data-processing inequality does not apply. This is not a limitation of the proof technique; it reflects the genuine logical dependence of the monotonicity on the Markov structure.

A proof of [H-suff] requires showing that the BFS depth-increment kernel $p(\sigma(\ell+1) \mid \sigma(\ell), c)$ is independent of the fibre label c. At the level of the BFS rank observable this is now established.

Definition 3.5 (BFS rank observable)**.** Write Heisenberg elements as (a, b, z) with horizontal part (a, b) and central part z, and let V_c be the representation space of the Weil block ρ_c. For a prime q, a fibre label c, a depth ℓ, and any fixed non-zero seed $v_0 \in V_c$, the BFS rank observable is

$$r_c(\ell) = \dim \operatorname{proj}_{V_c} \operatorname{span}\{\, \rho_c(a, b, z)\, v_0 : (a, b, z) \in B_\ell \,\}. \tag{3}$$

Proposition 3.6 (Structural sufficiency at the rank level)**.** *For every prime q, every fibre label c, and every depth ℓ,*

$$r_c(\ell) = \#\{(a, b) \in (\mathbb{Z}/q\mathbb{Z})^2 : (a, b, z) \in B_\ell \textit{ for some } z\}, \tag{4}$$

the size of the L^1-ball reached by the abelianised projection of B_ℓ. The right-hand side does not depend on c; hence $r_c(\ell)$ is identical for all fibre labels, $I(c; r(\ell)) \equiv 0$, and [H-suff] *holds at the rank level for every label, not only within conjugate pairs.*

Proof. By Schur's lemma the Heisenberg centre $(0, 0, z)$ acts on the irreducible block ρ_c as the scalar $e^{2\pi icz/q}$, so $\rho_c(a, b, z)v_0 = e^{2\pi icz/q} \rho_c(a, b, 0)v_0$. For fixed (a, b) all centre translates are therefore proportional; since ρ_c is unitary, $\rho_c(a, b, 0)v_0 \neq 0$ for the non-zero seed, so the coset contributes exactly one direction to V_c, irrespective of c. The rank (3) thus equals the number of distinct horizontal cosets (a, b) reached in B_ℓ, which is the abelianised footprint (4) and carries no dependence on the central character. Since $r_c(\ell)$ is the same deterministic function of ℓ for all c, it carries no information about the label, so $I(c; r(\ell)) = 0$ at every depth and the chain $c \to r(\ell) \to r(\ell')$ is trivially Markov. The underlying intra-coset structure is established in [9]. □

Corollary 3.7 (Rank-level composability and redundancy of [H-sg]). *On the pre-saturation window, where $\ell \mapsto r_c(\ell)$ is strictly increasing, the rank coarse-graining channel $G^r_{\ell'\leftarrow\ell} : r(\ell) \mapsto r(\ell')$ is single-valued and composes, $G^r_{\ell_3\leftarrow\ell_2} \circ G^r_{\ell_2\leftarrow\ell_1} = G^r_{\ell_3\leftarrow\ell_1}$ for all $\ell_1 < \ell_2 < \ell_3$, so* [H-sg] *holds at the rank level. Moreover* [H-sg] *is not needed at the rank level: since $I(c; r(\ell)) \equiv 0$ by Proposition 3.6, the inequality $I(c; r(\ell')) \leq I(c; r(\ell))$ holds for every pair $\ell' > \ell$ directly, with no step-by-step lifting.*

Proof. By Proposition 3.6 the value $r_c(\ell)$ is the abelianised L^1-ball count (4), strictly increasing in ℓ until the ball fills $(\mathbb{Z}/q\mathbb{Z})^2$. On this window ℓ is recovered from $r(\ell)$, so $G^r_{\ell'\leftarrow\ell}$ is the single-valued advance of the common deterministic trajectory; advancing from ℓ_1 to ℓ_2 and then to ℓ_3 reaches the same value as advancing directly to ℓ_3, which is the stated composition. The second statement is immediate from $I(c; r(\ell)) \equiv 0$ at every depth. □

Proposition 3.6 supersedes the earlier pair-level observation: the conjugation symmetry $\rho_c \cong \rho_{q-c}$, which gives $\sigma_c(\ell) = \sigma_{q-c}(\ell)$ within a conjugate pair, is the special case of label-blindness restricted to a single orbit. It is also the correct analytical replacement for the spectral route, which is closed: the global Weil-block spectra for c and ωc are generically distinct [9], so the capacity $\sigma_c(\ell)$, strictly finer than the spectrum, cannot be controlled spectrally. The remaining step to the full hypothesis concerns the capacity rather than the rank: $\sigma_c(\ell)$ differs from $r_c(\ell)$ by the Born–Infeld fingerprint of the admissibility pipeline [6, 7], and one must show that this fingerprint preserves the c-independence of the increment. By Proposition 3.6 the rank carries no fibre information, so the entire residual c-dependence of $\sigma_c(\ell)$ must enter through the fingerprint; this is the single open gap, recorded as open problem [O-1a′] (Section 6).

The full extent of [H-suff] — including cross-pair dynamics — has been tested numerically for $q \in \{61, 151, 211\}$; see Section 5.3.

4 The Fibre-Erasure Theorem

4.1 The information functional

Definition 4.1 (Residual fibre information). The *residual fibre information* at depth ℓ is

$$I(c; \sigma(\ell)) = H(c) - H(c \mid \sigma(\ell)) = H(\sigma(\ell)) - H(\sigma(\ell) \mid c), \tag{5}$$

where $H(\cdot)$ denotes Shannon entropy with respect to measure μ.

Remark 4.2 (Physical interpretation). $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$: the BFS profile carries no information about the microscopic fibre representative. This is the operational statement of fibre erasure under Π.

4.2 Main theorem

Theorem 4.3 (Fibre-Erasure Theorem)**.** *Assume hypotheses* [H-suff] *(Hypothesis 3.1) and* [H-sg] *(Hypothesis 2.5). Then for all $\ell' > \ell \geq 0$:*

$$I\big(c; \sigma(\ell')\big) \;\leq\; I\big(c; \sigma(\ell)\big). \tag{6}$$

In particular, $I(c; \sigma(\ell))$ is non-increasing in ℓ and converges to a limit $I_\infty \geq 0$.

Proof sketch. By [H-suff], the triple $(c, \sigma(\ell), \sigma(\ell'))$ forms a Markov chain (2). The data-processing inequality [10] applied to this chain gives (6) directly. Composability [H-sg] extends the inequality from $\ell' = \ell + 1$ to all $\ell' > \ell$. □

Corollary 4.4 (Fibre-erasure equilibrium condition)**.** *Under the hypotheses of Theorem 4.3, $I(c; \sigma(\ell)) = 0$ if and only if $\sigma(\ell) \perp c$. This is the* fibre-erasure equilibrium*: the coarse-graining has fully collapsed the fibre-label information. It is realised by the thermodynamic universal profile σ^* in the limit $q \to \infty$ (Remark 5.3).*

Explicit disclaimer. Theorem 4.3 is *not* a c-theorem. At the fibre-erasure equilibrium, $I_\infty = 0$ does not count any effective degrees of freedom. A c-theorem analog requires a separate counting functional; see open problem [O-2].

4.3 Information decomposition and programme connections

By the chain rule for entropy:

$$H\big(\sigma(\ell)\big) = H\big(\sigma(\ell) \mid c\big) + I\big(c; \sigma(\ell)\big). \tag{7}$$

Here $I(c; \sigma(\ell))$ is the fibre-erasure functional (Theorem 4.3), monotone non-increasing in ℓ; $H(\sigma(\ell) \mid c)$ is the intrinsic entropy of the observable conditioned on the fibre label, and is the natural candidate for a counting monotone [O-2].

Remark 4.5 (Connection to PTO and axis orthogonality). The temporal monotone $I(n) = \sigma_{\mathrm{pair}}(0) - \sigma_{\mathrm{pair}}(n)$ of [5] lives on the temporal axis n (Table 1). The present $I(c; \sigma(\ell))$ lives on the coarse-graining axis ℓ. They measure distinct phenomena on orthogonal axes and should not be conflated.

5 Numerical Proxies and Spectral Programme Connections

5.1 Inter-sector variance as proxy

Definition 5.1 (Inter-sector variance)**.** The *inter-sector variance* at depth ℓ is

$$\mathcal{V}(\ell) = \mathrm{Var}_c\big(\sigma_c(\ell)\big) = \mathbb{E}_c\big[\sigma_c(\ell)^2\big] - \big(\mathbb{E}_c[\sigma_c(\ell)]\big)^2. \tag{8}$$

$\mathcal{V}(\ell) = 0$ is a necessary condition for $I(c; \sigma(\ell)) = 0$, but $\mathcal{V}$ is *not* guaranteed to be monotone in ℓ. Non-monotone behaviour of $\mathcal{V}$ would not contradict Theorem 4.3; it would indicate that variance is an imperfect surrogate for $I(c; \sigma(\ell))$.

5.2 Triplet variance and [H-color]

For $q \equiv 1 \pmod 3$, the *triplet-restricted variance* is

$$\mathcal{V}_{\mathrm{triplet}}(\ell) = \mathrm{Var}\big(\sigma_c(\ell),\ \sigma_{\omega c}(\ell),\ \sigma_{\omega^2 c}(\ell)\big), \tag{9}$$

averaged over colour triplets.

Remark 5.2 ([H-color] as a fibre-erasure statement). Hypothesis [H-color] of [9, 8] asserts $\sigma_c(\ell) = \sigma_{\omega c}(\ell) = \sigma_{\omega^2 c}(\ell)$ for $q \equiv 1 \pmod 3$, which is precisely $\mathcal{V}_{\text{triplet}}(\ell) = 0$ and, under [H-suff], $I_{\text{triplet}}(c; \sigma(\ell)) = 0$. The fibre-erasure theorem provides the information-theoretic framework within which [H-color] is a special case: the colour label becomes unobservable at the effective level. The triplet-restricted variance is already computable from the $q \in \{61, 151, 211, 307\}$ runs of O32 [8].

5.3 Numerical evidence for [H-suff]

Hypothesis [H-suff] was tested numerically on the BFS capacity data from the O25 pipeline [6, 7] for $q \in \{61, 151, 211\}$. For each depth step $\ell \to \ell+1$, the k-nearest-neighbour regression residuals $e_c = \sigma_c(\ell+1) - \hat{f}(\sigma_c(\ell))$ are computed from all $q-1$ block-averaged mean trajectories, where $\hat{f}$ is estimated without reference to c. The Spearman correlation $|\rho(e_c, c)|$ is then computed and compared to a permutation null (500 shuffles of c), yielding a right-tail p-value and a standardised z-score. Under [H-suff], the residuals e_c should be independent of c, and $|\rho|$ should be statistically indistinguishable from zero at every depth.

q	depths tested	$\|\rho\|_{\max}$	z-score max	signif. ($p < 0.05$)
61	10	0.169	0.78	0/10
151	37	0.123	1.05	0/37
211	49	0.120	1.55	0/49

Table 2: Spearman test of conditional independence $\sigma(\ell+1) \perp c \mid \sigma(\ell)$ on BFS capacity data for $q \in \{61, 151, 211\}$. $|\rho|_{\max}$: maximum absolute Spearman correlation between residuals and the fibre label over all tested depth steps. z-score: standardised deviation from the permutation null. Significant: number of depth steps with $p < 0.05$; zero in all cases.

For all three primes and across all depth steps tested, $|\rho(e_c, c)|$ does not exceed 0.17 and no depth step yields $p < 0.05$. The results, shown in Figure 1, are consistent with the permutation null throughout: no systematic c-dependence in the step-by-step residuals is detected.

These results constitute numerical evidence that the BFS depth-increment dynamics does not depend on the fibre label c beyond the capacity profile $\sigma(\ell)$, consistent with [H-suff].

Limitations. The test is limited by the available data: the O25 pipeline [6, 7] stores only the mean BFS capacity profile averaged over M block samples per fibre label; per-block trajectories would enable a more powerful categorical test. No full O25 run is currently available for $q = 307$, though [H-color] at $q = 307$ has been confirmed directly by O32 [8].

[H-color] co-admissibility diagnostic. The leave-triplet-out η^2 test (panel C of Figure 1) detects strong within-triplet residual correlation at $q = 151$ ($z_{\max} = 3.07$) and moderately at $q = 211$ ($z_{\max} = 3.25$), consistent with the [H-color] co-admissibility established by O32 [8]: triplet members share equal σ profiles, giving equal residuals after leave-triplet-out regression. This is not evidence against [H-suff]; it is an incidental corroboration of [H-color]. The signal increases with better [H-color] quality: $q = 151$ has the smallest inter-triplet variance ratio ($R_{\text{var}} = 0.71 \times 10^{-3}$) of the tested primes, consistent with the strongest triplet signal.

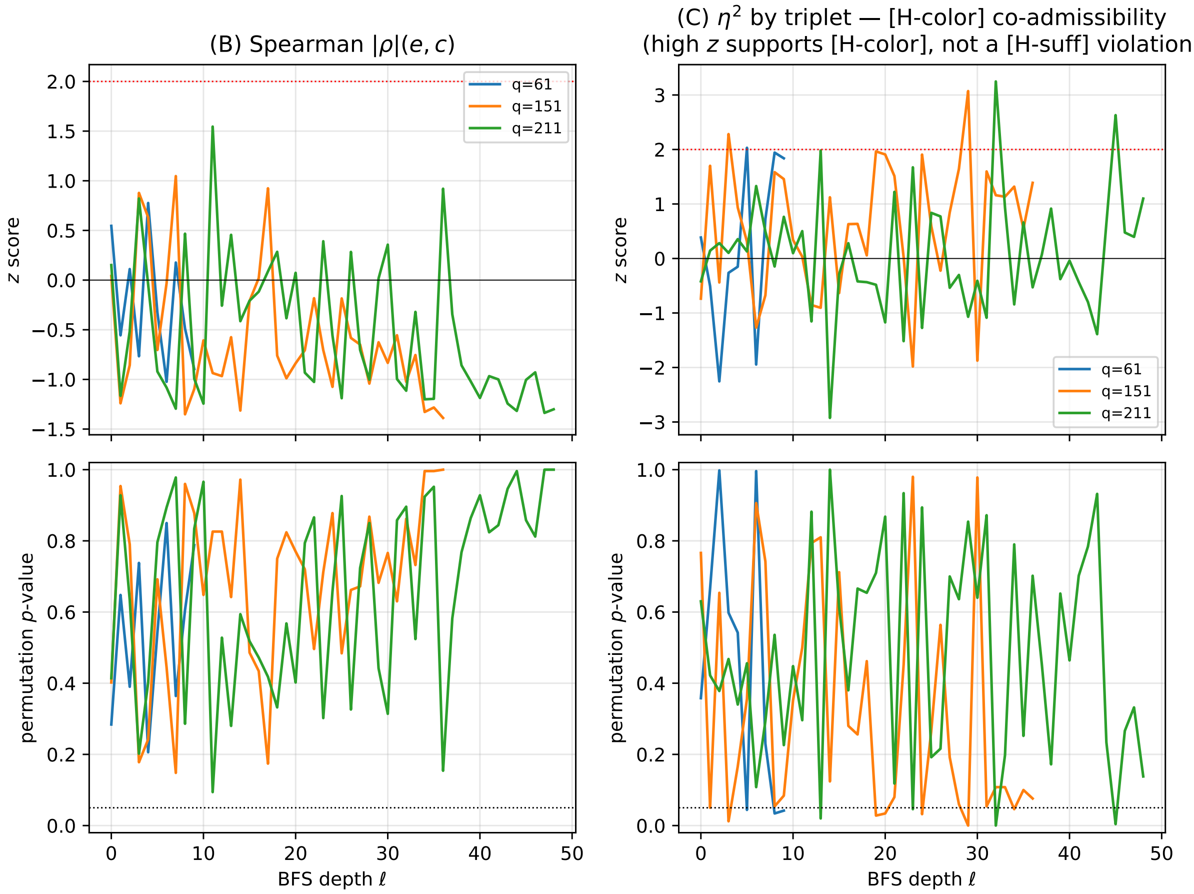

Figure 1: Numerical test of [H-suff] for $q \in \{61, 151, 211\}$. **Left panels (B):** Spearman $|\rho|(e_c, c)$ test (z-scores and p-values vs BFS depth ℓ). All z-scores lie below the $z = 2$ threshold (dashed red); no depth is significant. **Right panels (C):** leave-triplet-out η^2 test grouped by SU(3) colour triplet for $q \equiv 1 \pmod 3$; this measures [H-color] co-admissibility quality and does not test [H-suff] directly (see text).

5.4 Axis disambiguation: the fixed point is thermodynamic

The question of whether the fibre-erasure fixed point σ^* — the configuration where $I(c;\sigma(\ell)) = 0$ and fibre-label information is fully collapsed — is approached as a limit in the coarse-graining depth ℓ at fixed q (a Wilsonian fixed point of the semigroup $\{G_{\ell'\leftarrow\ell}\}$) or only in the thermodynamic limit $q \to \infty$ was investigated numerically on the BFS capacity data for $q \in \{61, 151, 211\}$. Two complementary diagnostics were computed.

Coefficient of variation. The scale-free inter-sector spread

$$\mathrm{CV}(\ell) = \frac{\mathrm{std}_c\big(\sigma_c(\ell)\big)}{\mathrm{mean}_c\big(\sigma_c(\ell)\big)} \tag{10}$$

equals $\mathrm{std}(A_c)/\mathrm{mean}(A_c)$, a depth-independent constant, if and only if profiles differ purely by amplitude: $\sigma_c(\ell) = A_c\,\sigma^*(\ell)$. Decrease of $\mathrm{CV}(\ell)$ to zero would indicate convergence to a common profile, i.e. a Wilsonian fixed point at fixed q. The data show $\mathrm{CV}(\ell)$ starting small at the fitting-window entry (0.045–0.079 across $q \in \{61, 151, 211\}$) and growing throughout the pre-saturation regime at all three primes (Figure 2, left panels); no plateau or decrease toward zero is observed. This rules out Wilsonian convergence at fixed q.

Amplitude-ratio test. For 20 random pairs (c, c'), the coefficient of variation of the ratio $\sigma_c(\ell)/\sigma_{c'}(\ell)$ over depth measures shape differences beyond amplitude scaling. Table 3 shows this quantity decreasing with q. The trend from $q = 61$ (0.156) to $q \in \{151, 211\}$ (0.065–0.062) is consistent with $\mathrm{O}(q^{-1/2})$ corrections from U1 universality [6, 7]: at larger q, profiles approach a common shape.

q	ratio-CV mean	ratio-CV std
61	0.156	0.069
151	0.065	0.021
211	0.062	0.025

Table 3: Amplitude-ratio CV over 20 random pairs (c, c'): coefficient of variation of $\sigma_c(\ell)/\sigma_{c'}(\ell)$ over depth in the pre-saturation regime. The decrease with q is consistent with $\mathrm{O}(q^{-1/2})$ shape corrections from U1 universality [6, 7].

Cross-q mean profile comparison. After normalising each mean profile by its value at the fitting-window start $\ell_0 = n_0$, the profiles for $q = 151$ and $q = 211$ are nearly identical, while $q = 61$ decays more rapidly (Figure 2, right panel). No collapse onto a single q-universal curve is observed at finite q.

These results establish that σ^* is a thermodynamic limit object: it is approached as $q \to \infty$ with $\mathrm{O}(q^{-1/2})$ amplitude corrections from U1 universality, not as a Wilsonian ℓ-limit at fixed q. At finite q, approximate fixed points $\sigma^*(q)$ exist with c-dependent amplitude corrections A_c that vanish as $q \to \infty$; the universal fixed point requires the continuum limit.

Remark 5.3 (Residual fibre information at finite q; terminology). Theorem 4.3 and Corollary 4.4 hold for any fixed q: their conclusion is that $I(c;\sigma(\ell))$ is non-increasing; they do not require the equilibrium condition to be reached. At finite q, a residual $I(c;\sigma(\ell)) > 0$ persists at all depths, reflecting the c-dependent amplitude corrections A_c.

A terminological precision is warranted. The *fibre-erasure fixed point* of Corollary 4.4 is the equilibrium condition $I(c;\sigma) = 0$ on the information functional — not a fixed point of the semigroup $\{G_{\ell'\leftarrow\ell}\}$, which has no non-trivial fixed point at finite q (all profiles eventually reach $\sigma_c(\ell) \to 0$ by BFS saturation). The universal profile σ^* realising this equilibrium condition is

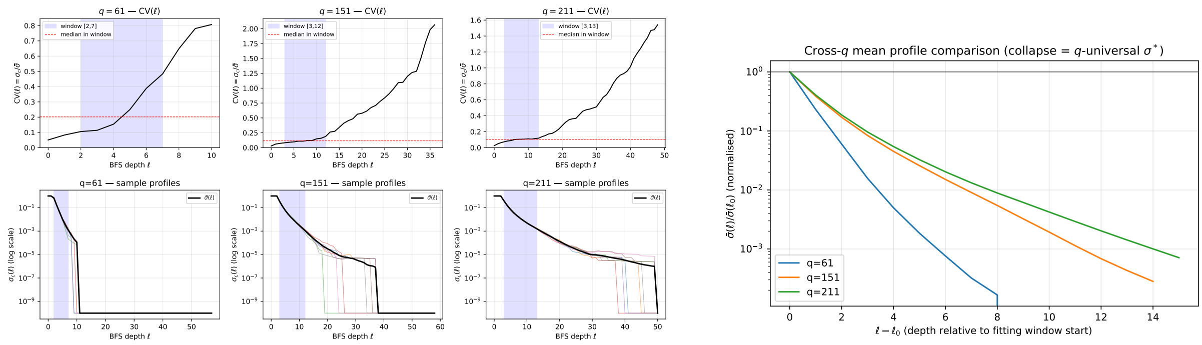

Figure 2: **Left:** CV(ℓ) (top row) and sample profiles on a log scale (bottom row) for $q \in \{61, 151, 211\}$. Blue band: fitting window $[n_0, n_1]$. CV(ℓ) grows monotonically throughout the pre-saturation regime; the sharp rise beyond the window is a saturation artefact (different c values reach $\sigma_c \to 0$ at different depths). No plateau or decrease toward zero is observed. **Right:** cross-q mean profiles normalised at $\ell_0 = n_0$. The $q = 151$ and $q = 211$ profiles are nearly identical; $q = 61$ decays faster. The absence of collapse confirms that σ^* requires $q \to \infty$.

therefore better understood as the *thermodynamic universal profile*: $\sigma^* = \lim_{q\to\infty} \sigma^*(q)$, where $\sigma^*(q)$ are approximate equilibria at finite q with $I(c; \sigma^*(q)) > 0$, approaching exact equilibrium only in the projective continuum limit $q \to \infty$ (Conjecture 4.7 of [2]).

5.5 Numerical evidence for [O-2]: conditional entropy monotonicity

The decomposition (7) identifies $H(\sigma(\ell) \mid c)$ as a candidate counting monotone (open problem [O-2]). A numerical proxy is available from the O25 pipeline extended with squared-profile accumulators: for each pair $(c, q-c)$ and BFS depth ℓ, the Bessel-corrected intra-block variance

$$\mathrm{Var}_{\mathrm{intra}}(\ell, c) = \frac{M}{M-1}\Big(\mathbb{E}_{\mathrm{block}}[\sigma_c(\ell)^2] - \mathbb{E}_{\mathrm{block}}[\sigma_c(\ell)]^2\Big) \tag{11}$$

estimates the spread of $\sigma_c(\ell)$ across the M independent block samples for fixed fibre label c. Under a Gaussian approximation, $H(\sigma(\ell) \mid c) \approx \mathrm{mean}_c\big[\frac{1}{2}\log(2\pi e\,\mathrm{Var}_{\mathrm{intra}}(\ell, c))\big]$, and its monotonicity reduces to that of the proxy

$$h_{\mathrm{intra}}(\ell) = \mathrm{mean}_c\big[\tfrac{1}{2}\log \mathrm{Var}_{\mathrm{intra}}(\ell, c)\big]. \tag{12}$$

Result for $q = 61$. On the BFS capacity data for $q = 61$ (30 conjugate pairs, $M = 20$ block samples, fitting window $[2, 7]$), $h_{\mathrm{intra}}(\ell)$ is perfectly monotone non-increasing over the pre-saturation regime: zero violations over five consecutive depth increments, linear-trend t-statistic -25.5, and a reduction of the intra-block standard deviation by a factor 0.024 (approximately $41\times$) over the window. Figure 3 shows $h_{\mathrm{intra}}(\ell)$, the between-c entropy proxy $h_{\mathrm{inter}}(\ell)$, and the normalised spread $\mathrm{CV}_{\mathrm{intra}}(\ell)$.

The normalised spread $\mathrm{CV}_{\mathrm{intra}}(\ell) = \mathrm{mean}_c\big[\mathrm{std}_{\mathrm{intra}}/\bar{\sigma}_c\big]$ increases over the same range ($0.46 \to 2.05$). This reflects the staggered-saturation artefact identified in Section 5.4: at late depths, some of the $M = 20$ blocks reach $\sigma_c = 0$ while others have not, inflating relative within-c variance. It is not a violation of [O-2]; the absolute proxy h_{intra} is the correct test statistic.

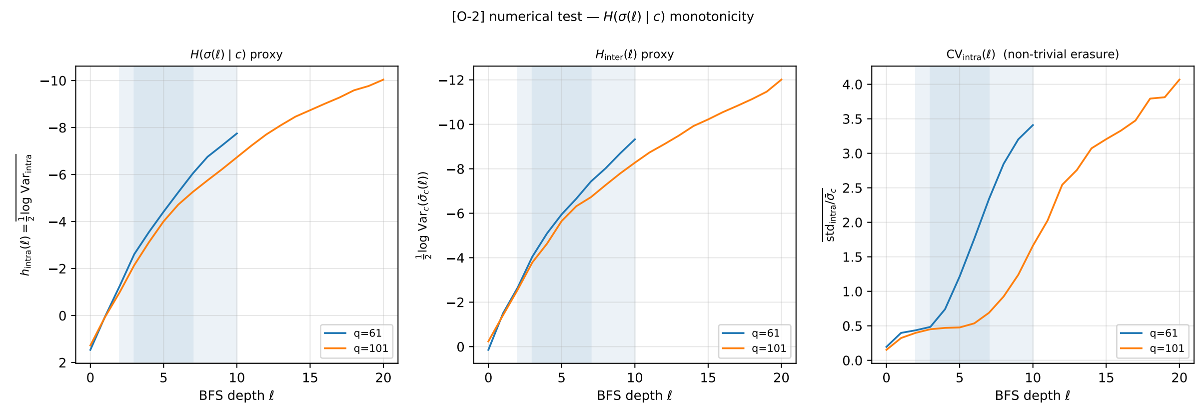

Figure 3: Numerical test of [O-2]: $H(\sigma(\ell) \mid c)$ monotonicity for $q \in \{61, 101\}$ (blue and orange respectively). **Left:** $h_{\mathrm{intra}}(\ell)$, proxy for $H(\sigma(\ell) \mid c)$; both primes are perfectly monotone non-increasing in the pre-saturation window (shaded band), with $t = -25.5$ at $q = 61$ and $t = -33.9$ at $q = 101$. The larger pre-saturation window at $q = 101$ directly reflects the larger q. **Centre:** $h_{\mathrm{inter}}(\ell)$, between-c entropy proxy. **Right:** $\mathrm{CV}_{\mathrm{intra}}(\ell)$; the increase at late depths is the staggered-saturation artefact (Section 5.4), not a violation of [O-2].

These results constitute preliminary numerical evidence that $H(\sigma(\ell) \mid c)$ is monotone non-increasing, consistent with [O-2].

Result for $q = 101$. On the BFS capacity data for $q = 101$ (50 conjugate pairs, $M = 20$ block samples, fitting window $[3, 10]$), $h_{\mathrm{intra}}(\ell)$ is perfectly monotone non-increasing: zero violations over six consecutive depth increments, linear-trend t-statistic -33.9, and a reduction of the intra-block standard deviation by a factor 0.021 (approximately $47\times$) over the window. The slope is $-1.009 \approx -1$, suggesting that $\log \mathrm{Var}_{\mathrm{intra}}$ decays near-linearly with BFS depth — a geometric concentration rate to track at larger q. $\mathrm{CV}_{\mathrm{intra}}$ increases ($0.46 \to 1.45$) for the same staggered-saturation reason as at $q = 61$ (Section 5.4). The signal is stronger than at $q = 61$ ($t = -33.9$ vs. -25.5, seven pre-saturation depths vs. five), confirming that the monotone behaviour is not a small-q artefact.

Status of the Gaussian proxy. The quantity $h_{\mathrm{intra}}(\ell)$ is not an exact Shannon entropy unless the within-c distribution of $\sigma_c(\ell)$ is Gaussian. It should be understood as a variance-based entropy proxy. Its monotonicity is nevertheless meaningful: any unimodal within-c distribution whose scale concentrates with ℓ has decreasing entropy under standard scale-sensitive estimators. Thus the present test probes concentration of the intrinsic observable spread, not a full non-parametric entropy estimate.

Counting requirement. A genuine counting monotone requires more than monotonicity alone. One must also show that $H(\sigma(\ell) \mid c)$ approaches a non-zero, stable, and interpretable value at the approximate equilibrium $\sigma^*(q)$, with a controlled q-dependence. If this limiting value vanished identically, the functional would not count effective degrees of freedom in the sense required by a c-theorem analog.

Role of the multi-q extension. With $q \in \{61, 101\}$ now confirmed, the remaining planned tests are $q \in \{151, 211\}$. They should determine whether the slope of h_{intra} stabilises near -1 across q and whether the decay rate scales compatibly with the decay of the fibre-erasure functional $I(c; \sigma(\ell))$. If the two terms of decomposition (7) decay with compatible exponents, the decomposition becomes an operational structural statement rather than a purely formal identity.

6 Rank-Level Closure and Open Problems

The rank-level components of the original problems [O-1] (analytical proof of [H-suff]) and [O-1b] (the semigroup hypothesis [H-sg]) are closed. Proposition 3.6 establishes [H-suff] for the BFS rank observable unconditionally: $r_c(\ell)$ is fibre-blind and $I(c; r(\ell)) \equiv 0$, so the chain $c \to r(\ell) \to r(\ell')$ is trivially Markov. Corollary 3.7 shows that, on the pre-saturation window relevant to the capacity analysis, the rank channel composes and [H-sg] is in fact not needed at the rank level, the all-pair inequality holding directly. The remaining problems below are genuinely open; in particular [O-1a′] isolates the single non-combinatorial residue of [O-1] and [O-1b], namely whether the Born–Infeld imprint preserves this fibre-blindness for the full capacity pointwise; at the averaged-exponent level it is supported (see [O-1a′]).

[O-1a′] Born–Infeld preservation of the rank kernel.

Since the rank carries no fibre information (Proposition 3.6), any residual c-dependence in the capacity $\sigma_c(\ell)$ of Theorem 4.3 must enter through the Born–Infeld fingerprint by which $\sigma_c(\ell)$ differs from $r_c(\ell)$ [6, 7]. This fingerprint is the dynamically active central phase $\psi_c(\gamma) = e^{2\pi i c\gamma/q}$ of the exact Weil projection, which couples the central coordinate to the dynamics and makes $\sigma_c(\ell)$ genuinely finer than the rank [3]. At the averaged-exponent level, capacity-level erasure is supported by the universality of the block decay exponent [3] and by the effective colour-triplet form [H-color]$_{\text{eff}}$ (equality of capacity exponents in the $q \to \infty$ limit) [8]. The open task is the pointwise statement, that the increment kernel factorises block by block,

$$p\big(\sigma(\ell+1) \mid \sigma(\ell), c\big) = p\big(\sigma(\ell+1) \mid \sigma(\ell)\big), \tag{13}$$

equivalently that the central phase contributes only the amplitude C_c, not the shape, of the block decay $\sigma_c(\ell) \sim C_c\,\ell^{-\delta}$. The finite-$q$ evidence is mixed but encouraging. The large inter-block variance of [3] is dominated by heterogeneous saturation depth across blocks — the staggered-saturation effect of Section 5.4 — so it conflates amplitude and timing with shape and does not establish shape non-universality. In the other direction, the colour covariance of [8] has numerical rank one: the triplet profiles are co-linear, differing by amplitude alone, which is positive evidence for the shape universality required here. Establishing it analytically would close [O-1a′] and the pointwise finite-q form of [H-color] together.

[O-1b] Born–Infeld composability of the capacity channel.

At the rank level, on the pre-saturation window relevant to the capacity analysis, the coarse-graining channels compose and [H-sg] is redundant (Corollary 3.7). What remains is whether the Born–Infeld fingerprint keeps $\sigma(\ell)$ a sufficient Markov state, so that the capacity channels compose as $G_{\ell_3 \leftarrow \ell_2} \circ G_{\ell_2 \leftarrow \ell_1} = G_{\ell_3 \leftarrow \ell_1}$ for all $\ell_1 < \ell_2 < \ell_3$. This is a facet of the same Born–Infeld property as [O-1a′]: both hold once the fingerprint depends only on the current coset support, since that makes $\sigma(\ell)$ a sufficient state, c-blind in its increment and Markov in ℓ. Without it, the data-processing inequality still yields step-by-step monotonicity from [O-1a′], but not the uniform statement of Theorem 4.3 for all $\ell' > \ell$ at the capacity level. As for [O-1a′], only the pointwise form is open; it stands or falls with the same shape-universality question.

[O-2] c-theorem analog via $H(\sigma(\ell))$.

The decomposition (7) identifies $H(\sigma(\ell) \mid c) = H(\sigma(\ell)) - I(c; \sigma(\ell))$ as a candidate counting monotone. Establish its monotonicity in ℓ and verify that its fixed-point value counts effective degrees of freedom. This would upgrade the fibre-erasure theorem to a genuine c-theorem analog. Numerical evidence for the monotonicity is now available for $q \in \{61, 101\}$ via the intra-block variance proxy (Section 5.5), with a consistent slope ≈ -1 at $q = 101$; cross-q confirmation at $q \in \{151, 211\}$ and an analytical proof remain open.

[O-3] Temporal causal structure and emergent light cone.
Define the temporal mutual information $I_{\text{temp}}(n,k) = I(U_n;\, U_{n+k})$ and the transfer entropy $T(U_n \to U_{n+k}) = I(U_{n+k};\, U_n \mid U_{n-1},\dots,U_0)$. Their decay in k would yield an operational definition of the emergent forward light cone on the temporal axis n, orthogonal to the present paper.

7 Conclusion

This paper establishes a monotone fibre-erasure property for the BFS coarse-graining hierarchy on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: admissible depth-increment coarse-graining cannot increase the fibre information $I(c;\sigma(\ell))$, so the label c becomes progressively less recoverable from the observable profile as the hierarchy deepens. The result is conditional on the structural sufficiency hypothesis [H-suff] at the full capacity level, and on the semigroup hypothesis [H-sg] for the uniform all-pair statement.

The central analytical contribution is the identification of a rigorous, unconditional partial closure. Proposition 3.6 shows that the BFS rank observable $r_c(\ell)$ equals the abelianised L^1-ball count, independent of c, so $I(c;r(\ell)) \equiv 0$ at every depth: fibre erasure is total at the rank level, for every label. Corollary 3.7 shows that the rank channel composes trivially on the pre-saturation window and that [H-sg] is not even needed at that level. Both results follow from a single mechanism (Schur's lemma forcing one direction per horizontal coset, irrespective of the central character c) established in [9]. This closes the combinatorial content of the original open problems [O-1] and [O-1b], and reduces the entire residual analytical content to a single question: does the dynamically active central phase of the exact Weil projection contribute only the block-decay amplitude C_c, not the shape $\ell^{-\delta}$? If so, both [O-1a′] and the pointwise finite-q form of [H-color] close at once.

The evidence is encouraging. At the averaged-exponent level, capacity-level erasure is supported by the universality of the block decay exponent [3] and the effective colour-triplet form [H-color]$_{\text{eff}}$ [8]. At the profile level, the colour covariance of [8] has numerical rank one at $q \in \{61, 151, 211, 307\}$: profiles within a colour triplet are co-linear, differing by amplitude alone. The large inter-block variance reported in [3] is dominated by heterogeneous saturation depth — a staggered-saturation timing effect (Section 5.4) — and does not constitute evidence of shape non-universality. For the candidate c-theorem analog [O-2], the intra-block conditional-entropy proxy $h_{\text{intra}}(\ell)$ is monotone non-increasing at $q \in \{61, 101\}$, with a consistent slope ≈ -1 suggesting geometric concentration of the within-c capacity spread. Tests at $q \in \{151, 211\}$ are planned.

Beyond the formal results, fibre erasure invites an interpretive reading. Because the rank-level erasure is exact and total rather than statistical, the loss of fibre information is structural: its combinatorial core is fixed by the Heisenberg geometry, and only the amplitude resides in the dynamical Born–Infeld fingerprint — a clean separation of the kinematics of erasure from its rate. The result also localises fibre identity: the abelianised footprint is label-blind, so all distinguishability between fibres is carried by the central phase, that is, by the very central extension that makes the group non-abelian. Coarse-graining erases the label by randomising this phase, an intrinsic decoherence in which the hierarchy itself plays the role of the environment. Read through the gauge interpretation of fibres, the fixed point $I(c;\sigma(\ell)) = 0$ (Corollary 4.4) is then the statement that the effective description is fibre-independent: gauge invariance of the emergent theory appears as the equilibrium of an irreversible coarse-graining flow, derived rather than imposed. In this reading the BFS hierarchy behaves as a discrete coarse-graining flow, and fibre erasure as its information-theoretic H-theorem, with [O-2] the corresponding candidate c-theorem analog.

In the Cosmochrony programme [5], the fibre-erasure property complements the capacity-exponent results of the O-series [6, 7] and the quantum-geometric reconstruction of the Q-series. The monotone reduction of fibre information under BFS coarse-graining is complementary to projective temporal ordering (PTO): it describes how fibre structure is irreversibly compressed along the resolution axis ℓ, while PTO describes the temporal ordering of admissible projected capacity along the temporal axis n.

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